

JESSICA YIN

ABOUT

I'm interested in how to leverage tactile sensing for dexterous robot manipulation. My work focuses on the intersection of sensor design, simulation, and robot control. I'm based in Seattle, WA.

EDUCATION

University of Pennsylvania, GRASP Lab

PhD in Mechanical Engineering

Thesis: Exploring Multimodal
Sensing Across the Stack for
Robot Manipulation

NSF GRFP Fellow

Advisers: Mark Yim & James Pikul

2020 - 2024

Carnegie Mellon University

BS in Mechanical Engineering
Minor in Physical Computing,
School of Computer Science

Dean's List (GPA 3.75+): Spring
2018 - Spring 2020

Advisers: Carmel Majidi & Tess
Hellebrekers

2016 - 2020

LINKS

-  [jessicayin.github.io](https://github.com/jessicayin)
-  jessicayin98@gmail.com
-  [linkedin.com/in/jessicayin](https://www.linkedin.com/in/jessicayin)
-  [Google Scholar Profile](#)

EXPERIENCE

RESEARCH SCIENTIST 12.2025 - Present
NVIDIA - Seattle, WA

- Lead tactile sensing research for dexterous hands on the Dex team in the Applied Deep Learning Research group

POSTDOCTORAL RESEARCHER 12.2024 - 12.2025
Meta FAIR - Redmond, WA

- Led full-stack robotics research: developed tactile glove hardware, data collection infrastructure, policy training and deployment pipelines, resulting in RAL submission
- Co-authored ICRA & RSS papers with two PhD interns about leveraging human data from smart glasses and tactile gloves
- Independently managed lab, equipment, and robots onsite

AI RESEARCH SCIENTIST INTERN 6.2023 - 1.2024
Meta FAIR - Redmond, WA

- Developed tractable, zero-shot tactile simulation for dexterous sim2real RL in IsaacGym, published at ICRA 2025
- Built Quest teleop stack for Allegro hand

RESEARCH SCIENTIST INTERN 5.2022 - 8.2022
Toyota Research Institute - Cambridge, MA

- Pioneered research about proximity and tactile sensing fusion to segment contact patches, published at ICRA 2025
- Co-inventor of 2 patents filed by TRI for sensor designs and algorithms

PHD STUDENT RESEARCHER 8.2020 - 11.2024
University of Pennsylvania - Philadelphia, PA

- Designed novel multimodal proximity and tactile sensor, awarded Best Student Paper at RoboSoft 2022
- Cross-disciplinary collaborations for robot applications with Penn Nursing and Penn Medicine, resulting in 2 publications
- Mentored 6 high school and undergraduate researchers

ROBOTICS RESEARCH INTERN 6.2019 - 8.2019
MIT Lincoln Laboratory - Lexington, MA

- Collaborated with 6 interns for autonomous drone and vehicle projects in the Rapid Prototyping Group

STUDENT RESEARCHER 8.2016 - 6.2020
Carnegie Mellon University - Pittsburgh, PA

- Co-authored 3 papers about tactile sensors for robotics and wearables in IROS, RoboSoft, Adv. Func. Materials
- Placed 3rd nationally in Society of Women Engineers Research Poster Competition
- CMU Undergraduate Excellence in Research Award (top 3 in Mechanical Engineering)

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SELECTED PUBLICATIONS

ROBOT LEARNING

OSMO: An Open-Source Tactile Glove for Human-to-Robot Skill Transfer

Jessica Yin, Haozhi Qi*, Youngsun Wi*, Sayantan Kundu, Mike Lambeta, William Yang, Changhao Wang, Tingfan Wu, Jitendra Malik, Tess Hellebrekers. Under review at RAL.

TactAlign: Human-to-Robot Policy Transfer via Tactile Alignment

Youngsun Wi, **Jessica Yin**, Elvis Xiang, Akash Sharma, Jitendra Malik, Mustafa Mukadam, Nima Fazeli, Tess Hellebrekers. Under review at RSS 2026.

Dexterity from Smart Lenses: Multi-Fingered Robot Manipulation with In-the-Wild Human Demonstrations

Irmak Guzey, Haozhi Qi, Julen Urain, Changhao Wang, **Jessica Yin**, Krishna Bodduluri, Mike Lambeta, Lerrel Pinto, Akshara Rai, Jitendra Malik, Tingfan Wu, Akash Sharma, Homanga Bharadhwaj. ICRA 2026.

Learning In-Hand Translation Using Tactile Skin with Shear and Normal Force Sensing

Jessica Yin, Haozhi Qi, Jitendra Malik, James Pikul, Mark Yim, Tess Hellebrekers. ICRA 2025.

TACTILE SENSORS AND SYSTEMS

Multimodal Proximity and Visuotactile Sensing with a Selectively Transmissive Soft Membrane

Jessica Yin, Gregory M. Campbell, James Pikul, Mark Yim. Robosoft 2022. *Best Student Paper Award*.

Proximity and Visuotactile Point Cloud Fusion for Contact Patches in Extreme Deformation

Jessica Yin, Paarth Shah, Naveen Kuppuswamy, Andrew Beaulieu, Avinash Uttamchandani, Alejandro Castro, James Pikul, and Russ Tedrake. ICRA 2025.

Wearable Soft Technologies for Haptic Sensing and Feedback (Review Paper)

Jessica Yin, Ronan Hinchet, Herbert Shea, Carmel Majidi. Advanced Functional Materials 2021.

Closing the Loop with Liquid-Metal Sensing Skin for Autonomous Soft Robot Gripping

Jessica Yin, Tess Hellebrekers, Carmel Majidi. RoboSoft 2020.

Liquid Metal-Microelectronics Integration for a Sensorized Soft Robot Skin

Tess Hellebrekers, Kadri Bugra Ozutemiz, **Jessica Yin**, Carmel Majidi. IROS 2018.

PATENTS

Multimodal Proximity and Visuotactile Sensing Through Transmissive Membrane

2023

US Patent App. 18/327,981 - University of Pennsylvania

Inventors: James Pikul, Mark Yim, **Jessica Yin**

Deformable sensors having an internal stereo depth sensor and an infrared sensor

2025

US Patent App. 19/316,042 - Toyota Research Institute

Inventors: Andrew Beaulieu, Naveen Kuppuswamy, **Jessica Yin**

Collapsible protrusions having a variable coefficient of friction

2025

US Patent App. 18/644,687 - Toyota Research Institute

Inventors: Andrew Beaulieu, Naveen Kuppuswamy, **Jessica Yin**

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ACTIVITIES

Workshop Co-Organizer	1.2026
CVPR 2026: Sense of Space - Multi-Sensory Modeling for Embodied Intelligence	
Invited Speaker, Panelist, Spotlight Talk	6.2025
RSS 2025 Workshop on Dexterous Manipulation, Workshop on Contact Dynamics	
Lead Teaching Assistant	8.2021, 1.2022, 1.2023
University of Pennsylvania: Statics, Junior Design, Scientific Computing	
Project Mentor	6.2021
Penn GRASP AI4All	

AWARDS AND HONORS

IEEE RoboSoft 2022 Best Student Paper Award	2022
NSF Graduate Research Fellowship (Computer Vision & Robotics)	2020-2024
CMU Undergraduate Excellence in Research Award	2020
CMU University Honors, College of Engineering Honors, Mechanical Engineering Dept. Honors	2020
CMU Mechanical Engineering Best Overall Capstone Project	2020
Society of Women Engineers Research Poster 3 rd Place, Top 10 National Finalist	2018
HackPrinceton Top 10 Finalist & 1517 Fund Award	2018
HackPrinceton Top 10 Finalist	2017
HackCMU Best Data Hack by Yahoo!	2016